Function Machines

Year 4 · **Homework Worksheet**

Worked Example

Function Machine Practice: A function machine doubles a number, then subtracts 4. What is the output when the input is 9?

Answer: 9 × 2 = 18; 18 − 4 = 14.

Function Basics (15 questions)

1. A function machine applies the rule ×7 to its input. What is the output when the input is 8?
2. A function machine applies the rule +12 to its input. What is the output when the input is 25?
3. A function machine applies the rule −9 to its input. What is the output when the input is 30?
4. A function machine applies the rule ÷6 to its input. What is the output when the input is 42? A) 6 B) 7 C) 8 D) 9 E) 10
5. A two-step function machine: input → +5 → ×2 → output. What is the output when the input is 6?
6. A two-step function machine: input → ×3 → +4 → output. What is the output when the input is 8? A) 24 B) 26 C) 28 D) 30 E) 32
7. A function machine applies the rule ×9 to its input. If the output is 54, what was the input?
8. A function machine applies the rule +18 to its input. If the output is 40, what was the input? A) 20 B) 21 C) 22 D) 23 E) 24
9. A function machine applies the rule −11 to its input. If the output is 9, what was the input?
10. A function machine applies the rule ÷7 to its input. If the output is 6, what was the input?
11. Select ALL that apply: a function machine applies the rule ×4 to each input. Which of these input→output pairs are correct? A) 3→12 B) 5→18 C) 6→24 D) 7→28 E) 9→32
12. A two-step function machine: input → ×5 → −6 → output. What is the output when the input is 4?
13. A two-step function machine: input → +9 → ÷3 → output. What is the output when the input is 12?
14. A function machine takes an input, trebles it, then subtracts 5. What is the output when the input is 6? A) 11 B) 12 C) 13 D) 14 E) 15
15. A function machine applies the rule −20 to its input. What is the output when the input is 20?

Applied Functions (15 questions)

1. A function machine: input → ×2 → +11 → output. What is the output when the input is 9?
2. A function machine: input → −7 → ×5 → output. What is the output when the input is 15? A) 35 B) 40 C) 45 D) 50 E) 55
3. A function machine: input → ÷4 → +9 → output. The input is 24. Find the output.
4. A machine takes an input, adds 6, then multiplies by 2. If the output is 28, what was the input? A) 7 B) 8 C) 9 D) 10 E) 11
5. A function machine: input → ×3 → −10 → output. The output is 11. Find the input.

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6. A function machine turns 4 into 15 and 7 into 24 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

7. A function machine: input → ×6 → output. What is the output when the input is 0.5?

8. Jamal puts a number into a machine. The machine doubles the number, then adds 7. The output is 23. What number did Jamal put in? A) 6 B) 7 C) 8 D) 9 E) 10

9. A function machine: input → +10 → output. What is the output when the input is −6?

10. Select ALL that apply: a function machine applies the rule "add 3, then multiply by 3" to inputs. Which input→output pairs are correct? A) 2→15 B) 4→20 C) 5→24 D) 6→28 E) 7→30

11. A function machine: input → ×4 → output. What is the output when the input is 3/4?

12. A function machine: input → ×0.5 → output. What is the output when the input is 19? A) 8.5 B) 9 C) 9.5 D) 10 E) 10.5

13. A machine trebles a number, then subtracts 6. If the result is 18, what was the original number?

14. A function machine: input → ÷5 → +8 → output. What is the output when the input is −10?

15. A function machine takes a number, adds 1/3 to it, then triples the result. What is the output when the input is 2/3?

Standard / Advanced (15 questions)

1. Tom thinks of a number. He doubles it, then adds 8, and ends up with 24. What number did Tom start with?

2. Maya triples a number then takes away 6, leaving 18. What was Maya's original number?

3. Leo takes a number, subtracts 4, then multiplies the result by 5. He gets 30. What number did Leo start with?

4. Hiring a handyman costs £20 just to show up, then £30 for every hour of work. One job's total bill was £110. How many hours of work does that cover?

5. Priya adds 6 to a number, then triples the result, getting 36. What was Priya's original number?

6. Take a number, add 5, multiply the result by 2, then subtract 6. Starting with 3, what do you end up with?

7. A fairground charges a £4 entry fee plus £3 per ride. Modelled as a function machine (input = number of rides, output = total cost), Alex pays £19 in total. How many rides did he go on? A) 3 B) 4 C) 5 D) 6 E) 7

8. Start with 5. Multiply it by 3, then add 7. What is the final answer?

9. A rule machine does '×3, then +2' to whatever you put in. If you put in 10, what comes out? A) 28 B) 30 C) 32 D) 34 E) 36

10. Halve a number, then add 9, and the result is 16. What was the number before you started?

11. A machine's two-step rule is 'add 4, then double the result'. Select ALL of the input→output pairs below that match this rule: A) 2→12 B) 3→14 C) 5→18 D) 6→22 E) 8→24

12. A taxi ride costs £5 to start, plus £2 for every kilometre travelled. If a ride costs £15 in total, how many kilometres was the journey?

13. A function machine applies the rule 'double the input, then add 7' to a starting number of 9. What is the output?

14. First, 8 is added to a number. Then that result is doubled and has 2 taken away. Starting from 4, what is the final answer? A) 20 B) 22 C) 24 D) 26 E) 28

15. A machine always does 'double the number, then take away 1'. If you run a number through this machine twice in a row (feeding the result back in), what do you end up with, starting from 3?

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